

PISA

PISA 2025 Technical Report

Draft version

13 Data Management Procedures

Introduction

In PISA, as in any international survey, standards and requirements for data collection guide the creation of an international database that enables valid within- and cross-country comparisons and inferences. For both Computer-Based Assessments (CBA) and Paper-Based Assessments (PBA), these standards and requirements are developed with three major goals in mind: consistency, precision, and generalisability. To support these goals, data collection and management procedures are applied consistently across all data to ensure data quality. As such, “data management” within the scope of the PISA survey refers to a collective set of procedures and tasks that each country performs to produce a verified, national database. With these procedures, national teams can avoid or at least minimise the potential for errors.

Although these international standards and requirements stipulate a collective agreement and mutual accountability among countries/economies and contractors, PISA is an international study that includes countries/economies with unique educational systems and cultural contexts. The PISA standards provide participants with the opportunity to adapt certain questions or procedures to local circumstances, or to add components specific to a particular national context. To address these national adaptations, a series of consultations was held with the national representatives of participating countries and economies to reflect their expectations in line with PISA 2025 technical standards. During these consultations, the data coding of the national adaptations to the instruments was discussed to ensure their recoding in a common international format. The guidelines for these data management consultations and recoding concerning national adaptations are described later in this chapter.

An important part of the data collection and management cycle is not only to control and adapt to the planned deviations from general standards and requirements, but also to control and account for the unplanned and/or unintended deviations that require further investigation by countries/ economies and contractors. Such deviations may, at times, compromise data quality and/or render data corrupt or unusable. For example, implementing non-standard testing procedures might, in turn, affect test performance (e.g. session timing, the administration of test materials, and support tools such as rulers and/or calculators). Sections of this chapter outline aspects of data management aimed at controlling planned deviations, preventing errors, and identifying and correcting errors when they arise.

Given the complexities of large-scale assessment administration and the compressed PISA timeline, it remains imperative to record and standardise data procedures as much as possible to align with national and international data management standards. These procedures are generalised to suit the individual cognitive test instruments and background questionnaire (BQ) instruments used in each participating country. As a result, a suite of products is provided to countries/economies to assist national teams in handling data management tasks in a standardised way, preparing the national database and minimising errors. These products include a data management manual, training sessions and videos, and a range of other materials, including the data management software.

This chapter summarises the data management quality control processes and procedures, as well as the collaborative efforts of contractors and countries/economies to produce a final database.

Data management at the international and national level

Data management at the international level

To ensure compliance with the PISA technical standards, the following procedures were implemented by ACER Data Management to ensure data quality:

- Developed standards, guidelines, and recommendations for data management
- Facilitated data trainings and provided training resources (e.g. training exercises, lessons, and resource guides)
- Provided support for national team queries throughout the data management lifecycle
- Enhanced data quality and verification procedures considering new context or situations during processing and cleaning data at the international and national level
- Prepared databases and reports for use by contractors, OECD, and the National Centres
- Prepared interim and final data products (e.g. Data Explorer, compendia files) for dissemination to National Centres, the OECD, and, eventually, the public

Ensuring compliance with technical standards also involved close collaboration with project partners. In PISA 2025, the ACER Data Management team worked closely with all consortium members to ensure all data capture and quality procedures were accurately executed.

Data management at the national level

As the standards for data collection and submission involve a series of technical requirements and guidelines, each participating country appointed a National Project Manager (NPM) to organise the survey data collection and management at the National Centre. NPMs were responsible for ensuring that all required tasks, especially those relating to the production of a quality national database, were carried out on schedule and in accordance with the specified international standards and quality targets. The NPM was responsible for supervising, organising and delegating the required data management tasks at the national level. In addition, as these data management tasks require more technical data analysis skills, NPMs were strongly recommended to appoint a National Data Manager (NDM) to complete all data-related tasks on time and to supervise support teams during data collection and data entry. These technical tasks for the NDM included, but were not limited to, data verification and validation with a series of consistency and validity checks.

The data management process and quality control

For the PISA 2025 Main Survey, data management activities spanned pre-assessment, during-assessment, and post-assessment phases for both Computer-Based Assessment (CBA) and Paper-Based Assessment (PBA) administrations. Collectively, these activities supported the production of complete and internally consistent national data files that met PISA 2025 Technical Standards and enabled subsequent processing, verification and delivery of analysis-ready data.

At the national level, National Centre teams coordinated required activities, typically under the supervision of the National Project Manager (NPM), with a National Data Manager (NDM) undertaking data-related tasks and liaising with the Data Management contractor. In PISA 2025, key data management deliverables and task status were coordinated and tracked through the PISA Connect Workflow Manager, which supported structured communication, submission tracking and follow-up across the data management lifecycle, including the exchange of validation packages, discrepancy documentation and reconciled outputs.

Pre-assessment preparation and sampling management

In the pre-assessment phase, within-school sampling and related preparation activities were undertaken using the nationally configured ACER Maple software. ACER Maple is ACER's software system for within-school sampling, participant tracking and data management. The system supports systematic random sampling, instrument allocation, tracking and data entry through controlled workflows with integrated validation checks. ACER Maple can be deployed in hybrid online/offline modes to support field operations in low-connectivity environments, with online deployment used for centralised data entry and monitoring. Role-based access controls, automated error detection and standardised reports support data integrity and quality assurance across the data management lifecycle.

ACER Maple was used to confirm each National Centre's within-school sample design and configuration, undertake within-school sampling, and allocate instruments across both modes of administration. Pre-assessment sampling data were submitted through ACER Maple at defined points before the assessment period (including an initial sample submission and, where applicable, a subsequent update). These submissions formed the basis for subsequent preparation activities, including downstream platform preparation for Computer-Based Assessment (CBA) administrations and instrument allocation, tracking configuration, and data-entry readiness for Paper-Based Assessment (PBA) administrations. The submission process also enabled early identification of discrepancies in participant identifiers and respondent counts before test administration.

For CBA countries/economies, pre-assessment preparation additionally included participant rostering. Rostering comprised transforming sampling data submitted by National Centres through ACER Maple into the files required by the assessment platform, and loading these data into TAO, ensuring that sampled students were assigned to the correct schools, test forms (batteries), and login credentials. ACER applied system-based checks following submission to verify data completeness and internal consistency before loading into the assessment platform. Both initial and subsequent rostering submissions followed the same process. They required a short lead time to complete platform synchronisation, which National Centre teams accounted for when scheduling assessment sessions, particularly where late sampling updates or re-sampling occurred.

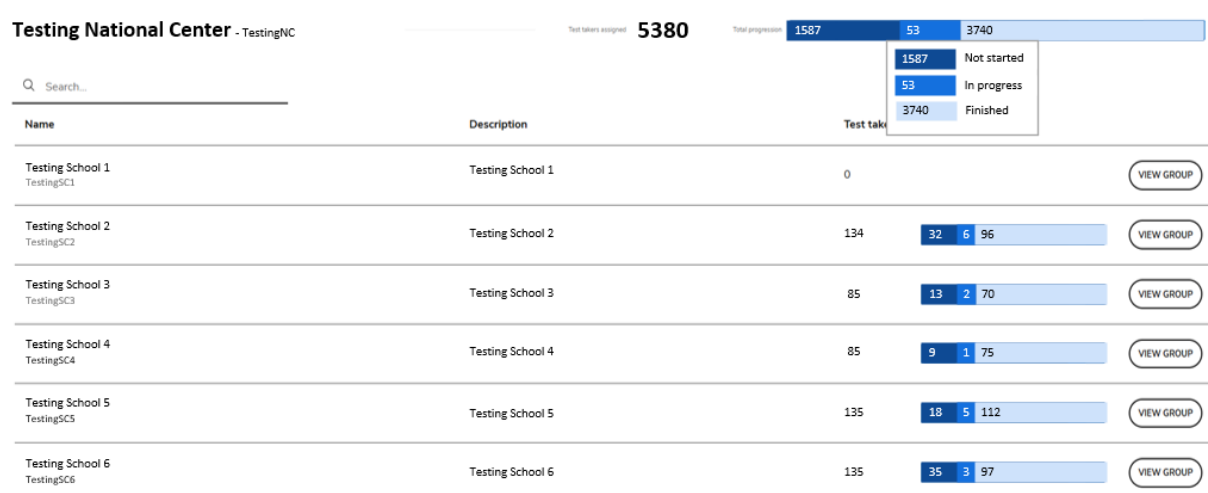
For PBA administrations, pre-assessment preparation focused on ensuring readiness for the material-based administration and subsequent data capture. This included confirming instrument requirements, establishing respondent-tracking arrangements, and preparing for data entry and reconciliation upon the return of physical materials. ACER Maple supported these activities by providing a single system for instrument allocation, configuration tracking, and the consolidation of pre-assessment information before test administration. For CBA administrations, participant rostering and assessment status information were available through the Assessment Platform Dashboard. Following sample submissions, National Centre teams verified that sampled participants were correctly represented in the platform and that counts and identifiers aligned with national tracking documentation across respondent types (students and, where applicable, teachers), noting that School Questionnaire respondents were represented separately in the platform. Any discrepancies identified through these comparisons (e.g. missing or duplicate identifiers, incorrect instrument allocation, unexpected respondent counts, or other mismatches between sampled participants and platform records) were documented and communicated to ACER Data Management for investigation and resolution.

Assessment-period tracking and monitoring

During the assessment period, National Centre teams monitored assessment progress and maintained tracking information, using mode-appropriate sources, to support accurate post-assessment reconciliation. For CBA administrations, platform status information from the Assessment Platform Dashboard was used to monitor student-level and school-level participation status, with test takers classified as not started, in progress, or finished. National Centres were asked to compare dashboard status information with returned

tracking forms to identify inconsistencies between recorded participation and platform status, including non-permitted combinations of participation codes and platform states. Where relevant, issues affecting the availability or completeness of platform data for subsequent activities (for example, sessions remaining in an 'in progress' state after completion) were flagged and followed up through established communication channels.

Figure 13.1: Assessment Platform Dashboard



For Paper-Based Assessment (PBA) administrations, returned test booklets and associated tracking forms served as the primary source of evidence of student participation during the assessment period, with participation status consolidated in ACER Maple as materials were received and checked. Unlike CBA administrations, participation information for PBA administrations became available only after physical materials were returned from schools and relevant tracking information was entered and verified, rather than being observable in real time during test administration. School participation and questionnaire completion information were maintained and updated in ACER Maple during the assessment period. For CBA countries/economies that used contingency (spare) login details, corresponding usage records were maintained to support later reconciliation.

Post-assessment reconciliation and preparation for validation

Immediately after the assessment period, National Centre teams undertook initial data receipt and checking activities to support subsequent validation and submission. These activities included consolidating assessment materials and reviewing tracking documentation for discrepancies before updating ACER Maple. For PBA countries/economies, returned paper instruments (test booklets and questionnaires) were checked against tracking information to support accurate data entry and reconciliation before validation, and to support the production of data-entry extracts used in Stage 1 validation described in the next section.

Post-assessment updates to sampling and tracking information were incorporated into ACER Maple as information was received from schools, and updated data were submitted through the designated post-assessment processes. These updates supported reconciliation of participation status, instrument allocation, and respondent tracking before the validation activities described below.

As part of post-assessment validation preparation, National Centre teams extracted from ACER Maple a standardised input for Stage 1 validation activities undertaken with the Main Survey Data Validation Tool,

including sampling and tracking information used to reconcile participant identifiers, participation status and questionnaire completion indicators against external sources.

Validation and discrepancy resolution

To support Main Survey timelines and quality assurance requirements, post-assessment validation and submission activities were organised into two stages. Stage 1 comprised national-level validation and reconciliation using the Main Survey Data Validation Tool, and Stage 2 comprised collaborative follow-up between National Centres and ACER Data Management to address remaining validation checks and resolve outstanding discrepancies before final processing.

In addition, National Centres' occupation coding activities were supported through cycle-specific occupation coding workbooks provided by ACER Data Management (and, where applicable, the OECD Risco tool). These workbooks supported the coding of open-ended occupation responses and associated quality control (including double-coding, as required). The resulting coded outputs were provided for use in subsequent validation and reconciliation checks.

Stage 1: National-level validation and reconciliation

For Stage 1, ACER Data Management provided National Centres with the Data Validation Tool package via the PISA Connect file exchange. For CBA administrations, the package included the platform extract used for validation, the validation executable, and associated configuration definitions. For PBA administrations, the package included the data-entry extract used for validation (e.g. the validation executable, and configuration definitions). These inputs, together with the standardised input for Stage 1 validation activities, supported the generation of discrepancy outputs for review, resolution and documentation.

The Data Validation Tool produced outputs to support review of consistency between ACER Maple tracking information and the corresponding platform (CBA) or data-entry (PBA) extracts. Outputs included a summary report, a machine-readable checks file (e.g. `discrepancy_checks.json`), and an associated discrepancy explanation form. The summary report and discrepancy explanation form set out the results of quality control checks that examined:

- participant identifiers and respondent counts in ACER Maple versus the corresponding platform (CBA) or data-entry (PBA) extracts
- recorded participation status information in ACER Maple versus the corresponding assessment status information available from the assessment platform (CBA) or from data-entry extracts (PBA)
- instrument allocation and questionnaire completion status information recorded in ACER Maple versus the corresponding respondent records in the relevant extracts for each respondent type (student, school, and, where applicable, teacher and parent)
- linkage consistency across respondent types, where applicable (for example, Parent Questionnaire records were correctly associated with student records)

Examples of discrepancies flagged by these checks include missing, duplicate or unexpected participant identifiers; inconsistencies between recorded participation status in ACER Maple and the assessment platform or data-entry records; non-permitted combinations of participation and platform status codes; and mismatches in instrument allocation or questionnaire completion across respondent types. In the summary report and discrepancy explanation form generated by the Data Validation Tool, the discrepancies were classified by check type and severity. For each flagged case, National Centre teams addressed the discrepancy either by updating sampling, tracking or participation information in Maple (and associated data-entry corrections for PBA where required) or by providing an explanation when a discrepancy could not be resolved within national source systems.

After addressing all discrepancies flagged by the quality control checks, the National Centres used the Data Validation Tool to produce updated, reconciled tracking data, which they then imported back into ACER Maple. They then compiled the materials for their Stage 1 submission, including the completed discrepancy explanation form and other documentation, in accordance with the cycle's submission processes. ACER Data Management used these artefacts to confirm resolution status, prioritise follow-up, and structure Stage 2 queries with National Centres.

Stage 2: ACER Data Management-led validation follow-up

Following completion of Stage 1 validation activities, National Centres submitted reconciled tracking data and associated validation outputs to ACER Data Management via the specified workflow steps and file exchange locations. ACER Data Management reviewed submissions and coordinated follow-up where additional clarification, corrections, or documentation were required.

Stage 2 involved structured follow-up between National Centres and ACER Data Management over the designated cleaning period to address remaining validation checks and discrepancies. This follow-up included, as applicable, verification of Parent Questionnaire participation and data entry, alignment between questionnaire data and occupation coding outputs, and resolution of issues identified through processing-derived indicators and other quality control outputs.

Where issues persisted, resolution involved iterative clarification and updates from National Centres, including follow-up with schools, as required, to verify information. National Centres remained available to respond to ACER Data Management queries during and immediately after the Stage 2 submission period, consistent with cycle timelines.

Across Stage 1 and Stage 2, sampling and tracking information maintained in ACER Maple served as the authoritative reference for participant eligibility, participation status and instrument allocation. They were validated against other relevant sources to support completeness and internal consistency before downstream processing.

After validation and reconciliation activities were completed and outstanding issues were resolved or documented, the data progressed to the next phase of processing. The procedures mirror the principle established in previous cycles that data quality assurance was an iterative, collaborative process between National Centres and international contractors, continuing well after initial data submission.

Data Processing and Cleaning

The data processing pipeline periodically retrieved newly extracted assessment data for countries/economies and begins by removing exact duplicate rows, excluding non-scored items such as introductory screens, correcting known item label errors, and classifying items into their respective types, including simple and complex multiple choice, computer-scored and human-coded open responses, and reading fluency tasks. For FLA, the same processing logic was applied with some domain-specific adjustments. Gap-fill and matching items were unpacked from compressed formats, while speaking items were not scored at this point but were checked to confirm that an audio recording existed. Response timings were calculated, and all supporting structures, including the codebook, speaking lookup table, and output schema, were formally validated before the data were saved.

All processed data accumulated for a country was consolidated, including both recent records and previously processed batches. Additional data cleaning steps were performed to ensure consistency and accuracy, including recoding negative response durations as timed-out cases and removing duplicate responses while retaining only the most recent instance. Items were ordered and assigned positional indices, and any trailing unanswered items beyond the last valid response were recoded as not reached rather than missing. Records flagged for deletion during stage 1 validation were excluded, and recorded login re-mappings from the stage 1 validation were corrected. Student counts were again verified before

saving to ensure no unintended changes had occurred. Effort thermometer responses were processed separately but followed the same deletion and recoding rules.

After processing, countries/ economies were prepared for the final output. Duplicate student logins in the registration data were excluded entirely. For each country, only the most recent test session per student was retained, and each test record was annotated with delivery characteristics such as whether it was taken in a virtual machine or standard environment, the form type, and whether the test followed a linear or adaptive structure. Student test data were then matched to Maple records, which added demographic and administrative information, including school identifiers, geographic classifications, grade level, date of birth, gender, special education needs status, population group, attendance indicators, and weighting variables used for analysis.

Items that had no valid responses within a country were recoded as not applicable, and timing and effort data were appended to each record. A standardised senate weight was calculated so that totals sum consistently across countries/ economies, alongside existing sampling weights. Final outputs were then generated in multiple formats suitable for analysis, including a long format with one row per student-item combination and wide formats where each student occupies a single row with item-level data represented across columns, produced both as scored datasets and raw response datasets. Once all countries/ economies had been processed, the data was combined into international files.

Throughout the pipeline, strict quality controls were applied to ensure data integrity. The primary check at every stage was the verification that the number of unique non-flagged student-test combinations remains unchanged from the beginning to the end of processing, with any discrepancy preventing the data from being saved. Additional validations confirmed that structural components, such as codebooks, lookup tables, and dataset schemas, met required standards and that no duplicate or malformed records were introduced. Further checks identify cases where human scoring was missing despite evidence that a student engaged with an item and ensure that all corrections applied to the data maintain internal consistency. Output datasets were also validated to confirm that the long and wide formats aligned correctly and remained internally consistent. At the same time, duplicate Maple records prevented a country from being processed, and unmatched student records were logged for review. Additional checks flagged unusual patterns such as students appearing in multiple test sessions on different dates, and separate validation processes can be run as needed to assess scoring reliability, summarise response distributions, verify consistency between outputs, and confirm that weighting variables aggregate correctly at the country level.

Cognitive Assessment Data Processing

Integration

After the initial data load and completion of early processing checks, the database entered the next phase of processing: Integration. During this integration phase, data structured within the country project database to support data collection was restructured to facilitate data cleaning. At the end of this step, a single dataset was produced for each respondent type: student, school, and teacher (where applicable). Additionally, Parent questionnaire data were merged with their child/student data.

Using this sampling information – captured in the Student Data File and Teacher Data File – extensive quality control checks were applied to the data in this phase. As a result of quality assurance checks, a data quality report was generated and delivered to countries/economies to resolve outstanding issues and inconsistencies. In this report, specific information was provided to countries/economies, including the name and description of the check, as well as student IDs for cases that proved inconsistent or incorrect with the check. These checks included (but were not limited to):

- Cognitive test (FORMCODE) variable was blank or not valid.

- Student was missing key data needed for sampling and processing.
- Student was not within the allowable age for the assessment.
- Student was not represented in the Sampling Data (Student Data File).
- Student was marked absent yet had a response record.
- Student's grade was lower than allowed.
- Student's assessment path misaligned with the multi-stage, adaptive design.
- A teacher was marked as a "non-participant," yet response data existed for that teacher.
- The data quality report was packaged along with a series of other quality control reports (i.e., harmonisation report and validation report, see "BQ Data Processing") for national team review. When reviewing the report, National Centre teams were asked to address flagged inconsistencies and correct data issues in the national database. National teams were instructed to complete the report review and data revision within a specific timeframe for resubmission to the Data Management contractor. Additionally, national teams documented all data revisions in the data quality report and returned the report to the Data Management contractor for review.
- After receiving the revised database and all documentation, the Data Management contractor repeated the pre-processing phase to ensure no new errors were reported and, if no issues or errors were found, the Data Management analyst re-executed the Integration step. As with the pre-processing consistency checks phase, the Integration step might have required several iterations and updates to country data if issues persisted and were not addressed by the National Centre. Frequently, one-on-one consultations were needed between the National Centre and the Data Management to resolve issues.

In addition to quality assurance reporting, a series of important data processing steps occurred during the Integration phase:

- **Item Cluster Analysis:** For data processing, it was often convenient to be able to disaggregate a single variable into a collection of variables. To this end, a respondent's single booklet number was generated as a collection of Boolean variables that signalled the item clusters to which the participant was exposed by design. Similarly, the individual item responses for a participant were interpreted and coded into a single variable representing the item clusters the participant appeared to have been presented with. An analysis was performed to identify inconsistencies between information in the student delivery system and that in the sampling design. Any discrepancies discovered were resolved by contacting the appropriate contractors.
- **Raw Response Data Capture:** In the case of Paper-Based Administration, individual student selections (e.g. A, B, C, D) to multiple-choice items were captured accurately. This was not necessarily true in the case of Computer-Based administrations. While the student delivery system captures a student's response, it does not capture data in a format suitable for distractor analysis. The web elements collected during the computer administration were therefore processed and converted into variables comparable to those in the Paper-Based administration.
- **Timing:** The student delivery system captured timing data for each screen viewed by the respondent. During the integration step, these timing variables were merged with the country database.
- **Process Data:** The student delivery system also produced log files where process data could be extracted for further analysis. Process data, including total response time, response time to first action, number of visits, and number of actions, was extracted using specialised tools and then verified by the Data Management contractor through a series of quality control checks. Such quality control checks identified inconsistencies or unreasonable situations (e.g. duplicate records, out-of-range values, system or operational issues or total unit duration exceeding item time). Once

inconsistent results were either resolved or explained, the data were provided to psychometric teams for further analysis.

- **SDS Post-processing:** Necessary changes in the student delivery system were sometimes detected after the platform was already in use. For example, a test item scored by the delivery system may have had an error in interpreting a correct response, which was corrected in post-processing. The delivery system's developers resolved these and other issues, and the Data Management contractor processed, issued, and merged the new scored response data.
- **Multi-Stage Adaptive Testing:** For the core domains (Science, Reading and Mathematics CBA), counts and percentages were produced for each country. Such counts identified the breakdown of each stage by performance, confirming that the student delivery platform's routing worked as expected during the assessment.

Scoring

After initial integration of the data, the next phase of data management processing involved parallel processes that occur with assessment data:

- Scoring of test responses captured in paper booklets
- Treatment of CBA human-coded items
- Additional checks of cognitive items

Scoring overview

The goal of the PISA assessment was to ensure comparability of the assessment results across countries/economies. As a result, scoring of the responses to the test items was a critical component of the data management processing. While scores for computer-based responses were generated automatically, no such scoring variables existed for Paper-Based components. This step in the process was dedicated to creating these variables and inserting the relevant student responses.

The scoring variables also served as a valuable data quality check. If any items appeared to function unexpectedly (i.e. too difficult, too easy, or unusually high missing rates), further investigation was conducted to determine whether a booklet printing or translation error occurred or whether systematic errors were introduced during administration, data loading, or data entry.

Each response was matched against the codebook and assigned a score code that distinguishes between incorrect responses, partially or fully correct responses, missing or unscorable responses, timed-out or anomalous cases, and items that will later be marked as not applicable. Human-scored responses from the ACER Marking System (AMS) were incorporated during processing, with priority given to the most recent available score. A series of targeted scoring corrections was applied at the item level, including rescoring open-response items using language-specific acceptable answer lists, recomputing scores where answer keys had been corrected, recalculating complex multiple-choice items when sub-parts were missing, and translating non-standard score codes into the standard format.

Once the Integration and Scoring steps were complete, the next phase of data cleaning involved validating the BQ data, i.e. harmonising national adaptations and verifying questionnaire response data.

Cognitive National Data Validation

For PISA 2025, student responses collected through the Computer-Based Assessment (CBA) platform were coded either automatically or by trained human coders. At this initial stage, these were referred to as raw codes, and they may be classified as valid or auxiliary. Valid codes represent the specific responses students provide, as recorded in the assessment system. For multiple-choice items, valid codes indicate the selected option. For example, multiple-choice items can have valid codes of 1, 2, 3, or 4 to indicate

that a student chose option A, B, C, or D, respectively. In instances where the number of multiple choice options was fewer or more than 4, the valid codes will vary accordingly.

In constructed response items, valid codes reflect scores assigned by human coders and include 0, 1, ..., m , where m corresponds to the maximum possible score for the item. Every valid code was included in the item analysis. The total number of valid cases for any given item matches the number of valid codes for that item stored in the database. Care should be taken when interpreting statistical analyses if any response category contains fewer than fifteen valid cases.

Missing responses occurring **before the last observed response** in a test session were treated as intentional omissions. These were coded as missing (code 9), scored as 0, and **included** in item analyses. Consecutive missing responses at the end of a test session were treated as not reached (code r) and were excluded from item-level analyses. Items not administered to a student by design were coded as *Not Applicable* (code 7) and excluded from analysis. These rules ensure that item statistics were computed only from responses that genuinely reflect item engagement, thereby protecting the validity of item parameter estimates.

PISA 2025 used Item Response Theory (IRT), implemented using ACER ConQuest, as the primary analytical tool for evaluating item performance. For each item, a set of national and international statistics was produced, enabling systematic detection of potential data quality issues. Trend item parameters were fixed to previously established estimates (e.g. PISA 2022), while new items were calibrated using international data to place them on the PISA reporting scale. A central validation mechanism in PISA 2025 was the flagging of items whose national performance deviated from international expectations. Items were flagged when they met predefined statistical criteria, including:

- **RMSD > 0.15**, indicating large deviations between observed and expected response probabilities
- **MNSQ (Infit) outside 0.7–1.3**, indicating poor fit to the IRT model
- **Negative point-biserial correlation for the key response**, or positive correlations for non-key responses with sufficient sample size
- **Low adjusted correlation (< 0.15)**, indicating weak discrimination
- **Non-monotonic ability ordering** for partial credit items

These thresholds were applied uniformly across countries/economies, reinforcing cross-national comparability and transparency. To support countries/economies in conducting a structured national review of flagged items, National Centres receive a National Report Database (Microsoft Access) and a pre-populated National Report Cognitive Feedback Form listing all flagged items. National Centres were required to:

1. Review statistical evidence for each flagged item.
2. Examine translations, adaptations, stimuli, and coding guides.
3. Decide on a preferred action (e.g. no action, delete item) with justification.
4. Document any newly identified errors.
5. Submit the completed feedback form within a defined timeframe.

Deleted items were treated as non-administered, ensuring that data quality decisions were formally recorded and consistently implemented in subsequent analyses.

Background Questionnaire (BQ) Data Processing

Harmonisation

Harmonisation, or harmonised variables

As mentioned earlier, although standardisation across countries/economies was needed, countries/economies had the opportunity to modify or adapt the BQ variable stems and response categories to reflect national-specificities or contexts. These adaptations were referred to as “national adaptations”. While able to capture country contexts, these adapted variables needed to be mapped into the corresponding international variable for cross-country comparison.

In the BQ instruments, national centres can introduce national adaptations by modifying the question stem and/or variable response category options to differ from the international version. This may involve the addition or deletion of a response option or a modification to the intent of the question stem or response option. For example, more response categories may have been added or deleted (e.g. a variable may have five response options/choices to the international question, but with the national adaptation, the variable may have been modified to only have four response options/choices as only 4 make sense for the country’s purposes); or perhaps two questions were merged. All of these were considered structural changes.

Harmonisation was the process of mapping the national response categories of a particular variable into the international response categories so they can be compared and analysed across countries/economies. See Box 13.1 for an example of a national adaptation that requires harmonisation.

Box 13.1. Example of a structural change

Addition of response categories to international questions – school type	
SC013Q01TA: International version	
Is your school a public or a private school? (Please select one response.) A public school (This is a school managed directly or indirectly by a public education authority, government agency, or governing board appointed by government or elected by public franchise.) A private school (This is a school managed directly or indirectly by a non-government organisation; e.g. a church, trade union, business, or other private institution.)	
The country may alter the stem to create a national adaptation and request information on the “type” of school, in addition to whether the school is public or private. This adaptation may result in addition of a third response category.	
SC013Q01TA: National version	
Is your school: (Please select one response.) Maintained by the local authority ([national example])? Maintained by central government ([national example])? An independent school?	
To ensure international comparison, for national variables where there was a structural change national centres are requested to provide the appropriate mapping of each response category to any of the international response categories.	

¹ International Standard Classification of Education

Not every nationally adapted variable required harmonisation, but for those that did, the Data Management team created harmonisation mappings for each country using R code. This code was implemented during the questionnaire data cleaning stage to handle these national variables during processing.

Overview of the workflow

To capture the appropriate adaptation and harmonisation, changes to variables by national teams were proposed during the translation and adaptation process. National adaptations for questionnaire variables were agreed upon with the ACER questionnaires team. These discussions regarding adaptations happened in the negotiation phase between the country and the ACER questionnaires team, as well as the translation verification contractor, before data submission to ACER. All changes and adaptations to questionnaire variables were captured in the Questionnaire Adaptation Sheet (QAS).

It was the role of the ACER questionnaires team to use the country’s QAS file to approve national adaptations as well as any corresponding harmonisation mapping. The Data Management team also collaborated with the ACER questionnaires team in developing the harmonisation code for use in the cleaning process. Throughout this process, it was the responsibility of the ACER questionnaires team, with the assistance of the translation and verification contractor, to ensure the QAS was complete and reflected the country’s intent and interpretation.

The correct implementation of harmonisation for national adaptations was a collaborative task that involved consultation with the national teams during the BQ cleaning process and the review of National datafiles. Requested updates or changes were documented in the cleaning reports and feedback forms to the National data files.

Validation

In addition to the Harmonisation step, the data cleaning and verification process involved the Data Management team executing a series of validation checks on the national BQ data. A report was generated and submitted to the National Centres for review and feedback.

Validation overview

In addition to nationally adapted variables, the Data Management teams collaborated with the ACER questionnaires team to develop a series of validation checks that were performed on the data following harmonisation.

Validation checks were a set of consistency checks that provided National Centres with more detail concerning extreme and/or inconsistent values in their data. Issues detected by these checks were displayed in a validation report, which was shared with National Centres to observe these inconsistencies and, if necessary, request modifications to the data based on administrative information or make improvements for the next cycle of PISA.

Generally, validation checks captured inconsistent student, school, and teacher responses. For example, these checks captured an inconsistency between the total number of years teaching (TC007Q02NA) and the number of years teaching at a particular school (TC007Q01NA); or an inconsistency in school data related to the total number of full-time teachers (SC018Q01TA_A) and the total number of full-time science teachers (SC400Q01DA_A). Throughout the PISA cycle, these validation checks often provided valuable feedback on data quality.

Treatment of inconsistent and extreme values in the PISA 2025 main survey data

Following the approach implemented in PISA 2015, 2018 and 2022 for extreme and/or inconsistent values within national data, the Data Management team, the ACER Questionnaires team, and the OECD agreed to implement specific range-restriction rules during data cleaning to address such values. These values would be invalidated across all country databases.

The following principles were observed in the special handling of these inconsistent and/or extreme values:

- In most cases where there was an inconsistency, the question considered ‘more difficult’ was invalidated since this was more likely to have been answered inaccurately (for example, a question that involved memory recall or cognitive evaluation by the respondent; or, if an inconsistency existed between age and seniority, the proposed rule may invalidate seniority, but keep “age.”).
- Apply stringent consistency and validity checks while computing derived variables. With this principle, the original values may be kept, while the values for the derived variable may have applied an “invalid” rule.

The specific range restriction rules for PISA 2025 are presented in Table 13.A.3 **Error! Reference source not found..**

Derived variables

The ACER Questionnaires team generated R code to create derived variables for implementation in the cleaning system at this step in the process. The code to create derived variables included routines for

calculating these variables, treating missing data appropriately, adding variable labels, etc. This code was based on the Main Survey (MS) Data Analysis Plan, which outlined the derived variables calculated from PISA MS data.

For all questions in the MS questionnaires, whether or not they served as a basis for derived variables, the international database contains item-level data obtained from the delivery platform. For any derived variables, whenever possible, they were specified to be consistent with previous PISA cycles. In terms of this alignment, the first choice was to align with PISA 2015 to enable comparisons on science-related variables. The second choice was alignment with PISA 2022. This aimed to strike a balance and stability across recent and future cycles. A list of PISA 2025 Main Survey non-item response theory (IRT) derived variables ("simple indices") is available in Annex Table 13.A.1.

As part of quality control, all derivations were verified by the ACER's Data Management team. Any updates or recoding made to the derived variable code were completed and documented. Data files were refreshed to implement any changes to the code or the variables.

Deliverables

After all data processing steps were complete and all updates to the data were made by national teams to resolve any issues or inconsistencies, the final phase of data processing included creating deliverable files for specific contractors (e.g. Westat for Sampling) and the National Centres. Each data file deliverable required a unique specification of variables along with their designated ordering within the file.

In addition to the generation of files for contractors and National Centre use, the 'deliverables' step in the cleaning process contained critical additions to the data, such as the addition of proxy scores, plausible values, BQ simple derived indices, and sampling weights (student and school).

To produce these customised files for specific users including National Centres and the OECD Secretariat at specific project phases, each deliverable required a separate series of checks and reviews to ensure all data was handled appropriately and all values were populated as expected.

Preparing files for public use and analysis

To prepare for the public release of PISA 2025 Main Survey data, the ACER Data Management team provided SPSS data files (.sav) to National Centres and the OECD Secretariat in batch deliveries at various review points throughout the Main Survey cycle. With the initial data deliveries of the Main Survey, the data files included preliminary sampling weights and proxy proficiency scores for analysis. These data were later updated to include final sampling weights, plausible values, and questionnaire scaled indices.

During each of these phases of delivery, National Centres reviewed these data files and provided the ACER Data Management team with any comments and/or revisions to the data.

The following data files were delivered:

- The "Student" combined data file contained all student responses to the test items (raw and scored), BQ items, and optional questionnaire items such as Parent Questionnaire, Information and Computer Technology Literacy Familiarity (ICT) Questionnaire. These files included all raw variables, questionnaire-derived and scaled indices, student weights, replicate weights, and plausible values.
- The Foreign Language Assessment (FLA) Student combined data file contained all student responses to the test items (raw and scored), BQ items, and optional questionnaire items for the FLA Parent Questionnaire. These files included all raw variables, questionnaire-derived and scaled indices, student weights, replicate weights, and plausible values.

- The “School” data file contained all response data collected with the School Questionnaires (including the FLA school questionnaire). These files included all raw responses, school-level base weights, questionnaire indices, and other derived variables.
- The Teacher data file contained response data from the Teacher Questionnaire and the FLA Teacher Questionnaire. These files included all raw responses, questionnaire indices, derived variables, and teacher weights.
- The Masked international database combined the data from all participating countries/ economies. To preserve country anonymity in this file, key identifying variables were masked in accordance with specific guidelines from the OECD Secretariat, including the issuance of alternate codes or special handling for country identifiers.
- The preliminary, national version of the Public Use File (PUF) was produced toward the end of the PISA 2025 Main Survey. It provided the National Centre with the opportunity to review its data before the final public release. International-level recoding and country-requested variable suppressions were implemented to these data to strengthen data protection. Annex Table 13.A.4 provides further information on recoding at the international level, as well as on the suppression of variables at the national level.
- In addition to these data files, the Data Analysis team produced a series of analysis reports, which were delivered by the Data Management contractor to National Centres for quality control, data validation, and further national analyses. These reports were also used to evaluate the plausibility of distributions of background characteristics and the performance results by subgroup, especially the extent to which they agree with expectations based on external or historical information. These reports included:
 - BQ National Report: A report containing frequencies of categorical variables and summary statistics for numeric variables from the country’s BQs. To aid countries/ economies in reviewing their BQ variables for potential translation or coding errors, this report included flags for missing data rates above 5%.
 - Codebook Descriptives Report: A report that includes frequencies and percentages for all variables that employ a value scheme for cognitive and questionnaire variables, as well as those that had been derived and/or added during data cleaning and includes descriptive statistics for all variables.
 - Cognitive Summary Analysis Reports: A comprehensive report that included a series of key statistics and flags across item analysis (IA), coding reliability, and item response theory (IRT) reports to identify items that, based on the empirical data, were most likely to require careful review and feedback by national teams.
 - Item Analysis Reports: A set of reports that provided summary information about the response types given by the respondents to the cognitive items. They contained, for each country, various statistics (e.g. counts, percentages, mean cluster scores) on students' choices for multiple-choice items, or the per cent of individuals receiving each score in the scoring guide for the constructed-response items.

The Public Use File - Included Records

When preparing for the final public use file (PUF), the following records were included in the database:

Student files

- Includes one record per respondent¹ that met the international target population definition and that passed validation, adjudication, and weighting.

School files

- Includes one record per participating school – specifically, one record for any school with a student included in the PISA sample, regardless of whether the school returned the School Questionnaire.

Teacher files

- Include one record for each teacher that met the international target population definition and that passed validation, adjudication, and weighting².

Foreign Language Assessment (FLA) student files

- One record per student respondent that met the FLA target population definition and that passed validation, adjudication, and weighting; and that responded to the FLA cognitive forms.

Categorising missing data

Within the data files, the coding of the data distinguishes between six different types of missing data:

1. System Missing – used to indicate that no data was captured, or data loss.
2. No Response – used to indicate the respondent had an opportunity to answer the question but did not respond. For the simple derived indices, it was often used as an indicator for all different types of missing data.
3. Invalid – used to indicate that the response was not appropriate or contradicted a prior response, e.g. the response to a question asking for a percentage was greater than 100 or contained more than one choice for an exclusive choice type of question (PBA only).
4. Not Applicable – used to indicate in the questionnaire that the respondent was not presented the question according to the survey design, ended the questionnaire early or could not be determined due to a printing problem or torn booklet. In the cognitive data, an item or question was not presented to a respondent, for example, by design or because there was a technical error.
5. Valid Skip – used in the questionnaire data to indicate that the respondent was redirected based on their answer to a previous question.
6. Not Reached – used in the cognitive scored variables to indicate that the respondent was unlikely to have seen the item or question at the end of a test form.

Data management and confidentiality, variable suppressions

During the PISA 2025 cycle, some countries/economies' regulations and laws restricted the sharing of certain data with other countries/economies. The key goal of such disclosure control was to prevent the accidental or intentional identification of individuals in the release of data. However, suppressing information or reducing detail could affect the analytical utility of the data. Therefore, both goals must be carefully balanced. As a general directive for PISA 2025, the PISA Governing Board requested to strengthen data protection, while ensuring the Public Use Files (PUFs) available with the largest possible set of information at the highest level of disaggregation, so as to enable researchers to access PISA data in meaningful ways. In response to this request, international-level recoding and country-requested suppressions were applied in the preparation of the PUFs.

International-level recoding

The OECD Secretariat conducted an international reidentification risk assessment. This risk assessment was used to identify variables with high granularity and possible risk of re-identification of students,

schools and teachers. Based on the results of the risk assessment, a list of international recoding has been applied to a number of variables to reduce risk of reidentification while keeping international comparability. The following recoding have been applied:

- **Student Dataset:**
 - o **ST021Q01TA** (age of arrival in a host country) is recoded using the PISA 2025 International Report into three categories: **0-5 years**, **6-11 years** and **12 years or more**.
 - o **OCOP1** (parent/guardian 1 occupation) and **OCOP2** (parent/guardian 2 occupation) are recoded by reducing the occupational coding from **4 digits to 2 digits**.
- **School Dataset:**
 - o **SC002** (the number of boys and girls) is recoded to retain only **the integer part of the percentage of boys/girls in the school**.
 - o The following continuous count variables: **SC064Q** (parental involvement), **SC004Q** (digital devices), **SC016Q** (school funding), **SC018Q** (the number of teachers), **SC400Q** (the number of science teachers), **SCHSIZE** (school size), **SC211Q** (students with various characteristics), **SC168Q** (the number of non-teaching staff), **SCRATIO** (Student-science teacher ratio), **STRATIO** (student-teacher ratio), **TOTAT** (total number of teacher), **TOTSCIE** (total number of science teacher), **TOTSTAFF** (total number of non-teaching staff) are recoded into **international quartiles**, defining a fixed four-group distribution in which each category represents approximately **25% of students internationally**.
 - o The following ratio and proportion variables: **PROADMIN** (school administrative personnel), **PROATCE** (fully certified teachers), **PROFEMST** (female science teachers), **PROMGMT** (school management personnel), **PROPAT6** (teachers with ISCED6), **PROPAT7** (teachers ISCED7), **PROPAT8** (teachers ISCED8), **PROPSIE** (science teachers), **PROPSUPP** (personnel for pedagogical support), **PROSTCE** (fully certified science teachers), **PROWBST** (staff focused on well-being) are recoded into **five fixed categories between 0 and 1**, using intervals of **0.2**.
- **Teacher Dataset:**
 - o **TC002Q** (teacher age) is recoded into four age groups: **30 or below**, **31–39**, **40–49**, and **50 or above**.
 - o **TC007Q** (years of experiences), **TC216Q** (time spent various activities), **PROPWORK** (proportion of working years at this school), and **TCWKLOAD** (teacher workload) are recoded into **international quartiles**, using the internationally fixed threshold, following the same approach as the one applied for school continuous variables.

Country-requested suppressions and recoding

Each country was required to provide early notification of any rules affecting the disclosure and sharing of PISA sampling, operational or response data. Furthermore, each country was responsible for implementing any additional confidentiality measures in the database before delivery to the Consortium. Most importantly, any confidentiality edits that changed the response values had to be applied before submitting data to ensure identical values during processing, cleaning and analysis. The country implemented all other measures by editing individual data cells.

Country suppression requests generally involved specific variables that violate the confidentiality and anonymity of student, school, and/or teacher data in their national contexts. To suppress data for the public use files, an invalid code was applied during the final step of data file creation in the cleaning system³. A listing of suppressions and recoding at the country variable-level is in Table 13.A.4.

Notes

1. To be considered a “respondent” for weighting, the student must have met the definition for a respondent for at least one domain.

Students were deemed respondents for a cognitive domain if they gave:

- at least five responses to the cognitive assessment in one domain, **or**
- at least one response to a cognitive item in that domain ***and*** questionnaire items which contribute to the socio-economic index.

2. Teachers who were absent, excluded, or refused to participate in the session may be marked as a “non-participant.”
3. PISA national participants also had the opportunity to request a withdrawal of data earlier in the process, before accessing confidential international data files. These requests were managed by the OECD and implemented by the Data Management contractor. The withdrawal of data involves removing data (e.g. records from specific regions) from data files and reports (including public-use files) for country-specific reasons. The request to withdraw data required thorough discussion with the OECD and approval.

Annex 13.A. Additional Data Management Items

List of Tables in Annex 13.A

Table 13.A.1	PISA Non-IRT Derived Variables Code
Table 13.A.2	PISA Variable Naming Convention
Table 13.A.3	PISA 2025 Range Restrict Codes
Table 13.A.4	PISA Country Variable Suppressions and Recoding

Note: See the Excel file *13-AnnexTables.xlsx* for each table listed above.